

Luca Ciampaglia

Mechanical Engineer Electrical Engineer Robotics Bioinspired Design Biomimetics

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PROFESSIONAL SUMMARY

Over 3 years of research experience in electrical engineering, mechanical engineering, robotics, and mechanical design. Career focuses on design and fabrication of complex mechanical systems, with CAD and simulation expertise. Research interests in bioinspired design and biomimetic systems. 3 years of leadership experience of a multidisciplinary team in competitive engineering setting. Multiple preprint publications currently under review.

EDUCATION

Master of Science in Robotics, Mechanical Engineering

Northeastern University, Boston, MA

Graduated: 2026

Bachelor of Science in Electrical Engineering

Case Western Reserve University, Cleveland, OH

Graduated: 2024

WORK EXPERIENCE

Graduate Researcher

Northeastern University, Boston, MA

September 2024 – April 2026

- Researched robotics and mechanical design in the Silicon Synapse Lab under Dr. Alireza Ramezani
- Mechanical design of the biomimetic flapping wing robot Aerobat, regulation of flapping gait and thrust via actuator design and integration in wing structure
- End-to-end completion of robotics mechanical engineering research project Aerobat Delta
- Rapid fabrication and prototyping of compliant joints and mechanisms
- Complex mechanism design and fabrication, experimental testing and validation
- Defended and published thesis for MS degree, second publication under review

Undergraduate Researcher

Case Western Reserve University, Cleveland, OH

June 2023 – May 2024

- Researched robotics, electrical and mechanical engineering in the Soft Machines and Electronics Lab under Dr. Changyong Cao
- Mechanical design and fabrication of biomimetic aquatic robot, Finite Element Analysis in ANSYS
- Pneumatic system assembly and setup, control PCB fabrication
- Custom Python simulation coding for soft robotics and deformable elastic geometry
- Hybrid tendon cable tension and pneumatic soft robot control

Case Aeronautics Team Administration

Case Western Reserve University, Cleveland, OH

May 2021 – May 2024

- President- management of multidisciplinary engineering team collaboration, leadership of 30-person engineering team
- Risk Manager- management of component and system testing, team safety and coordination, event oversight
- Technical report writing and delivery, team budget oversight, SOLIDWORKS CAD and aeronautical engineering

SKILLS

- Dassault Systèmes SOLIDWORKS CSWA and PDM
- Python, MATLAB, C++, Java, Simulink, Simscape, Git, ROS, Gazebo, ANSYS

REFERENCES

Alireza Ramezani, PhD *Silicon Synapse Lab*

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siliconsynapse.sites.northeastern.edu

Northeastern University, Boston, MA

Changyong Cao, PhD *Soft Machines and Electronics Lab*

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